

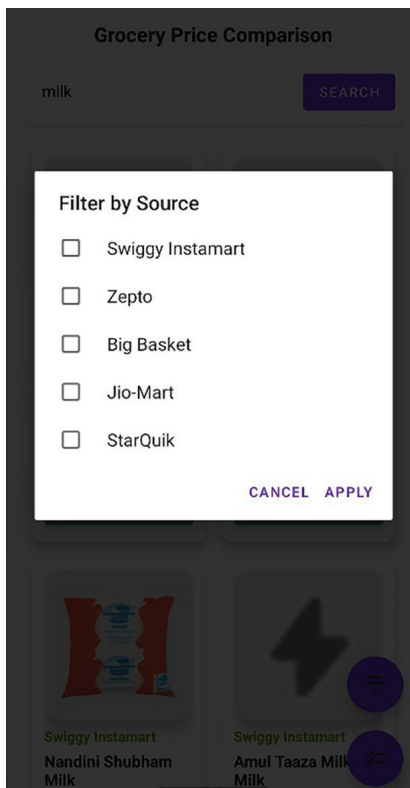


Figure 8: Filtering options for source selection in a user interface allowing users to refine search

combined with thoughtful design, can address practical consumer challenges while maintaining flexibility, scalability, and ethical integrity.

What still needs to be done

Although the system achieves its core objectives, there are several areas where functionality can be expanded. These enhancements span technical scalability, user experience, and system performance.

- **Improved scalability:** Asynchronous scraping using task queues like Celery or Kafka can help handle multiple platform requests in parallel and lower response latency.
- **Historical pricing and insights:** Adding backend support for logging historical price data would enable trend analysis and allow users to make more informed, data driven decisions.
- **User personalisation:** Features such as account creation, saved searches,

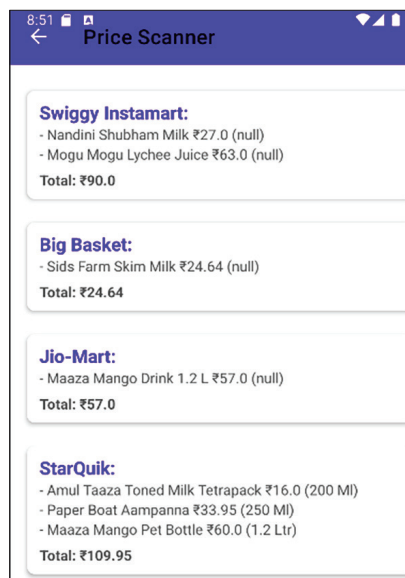


Figure 9: Comparison results displayed with price aggregation and sorting for easy decision making

price drop notifications, and delivery preferences could provide a more tailored experience.

- **ML-based recommendations:** Machine learning can be introduced

to suggest cost-effective alternatives, identify volume based savings, or rank products by price per unit.

- **Enhanced accessibility:** Improving support for screen readers, voice input, and keyboard navigation would make the app more inclusive for users with different accessibility needs.
- **Caching and scheduled updates:** Integrating intelligent caching and background refresh schedules would reduce scraping overhead and improve app responsiveness.
- **Domain expansion:** The platform could be extended to support food delivery and cab fare comparison, reusing the existing modular microservices framework to deploy new scraping containers with minimal overhead.

These additions would not only improve performance and user engagement but also strengthen the system's readiness for larger-scale deployment and broader use cases. **END** 🐧

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